18 Study Design & Statistical Claims

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RULE · Random sample (tasodifiy tanlanma) → generalize

Random sample → conclusion applies to the whole population. Convenience sample → describes only itself, no matter how large.

Size improves precision; only randomness improves representativeness.

RULE · Random assignment → causation

Random assignment to groups → "X causes Y." Observational study (no assignment) → "X is associated with Y" only – never causation.

A lurking variable can drive both at once – e.g. ice-cream sales and drownings both rise with hot weather, not with each other.

RULE · What can this study prove?

Ask both questions – they're independent:

Sample random?	Assignment random?	Can claim
Yes	Yes	Causation, whole population (rare)
Yes	No	Correlation only, whole population
No	Yes	Causation, this group only
No	No	Nothing reliable